Machine Learning Introduction 3

Neural Networks

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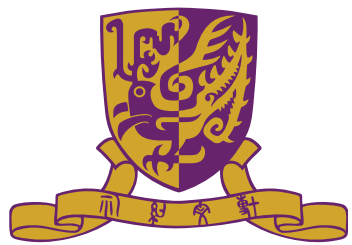

Agenda

The Linear Models We Studied and Limitations

We have mainly studied linear regression and linear classification.

- Least squares (linear regression), perception, logistic regression, and SVM. They are all linear supervised machine learning models.
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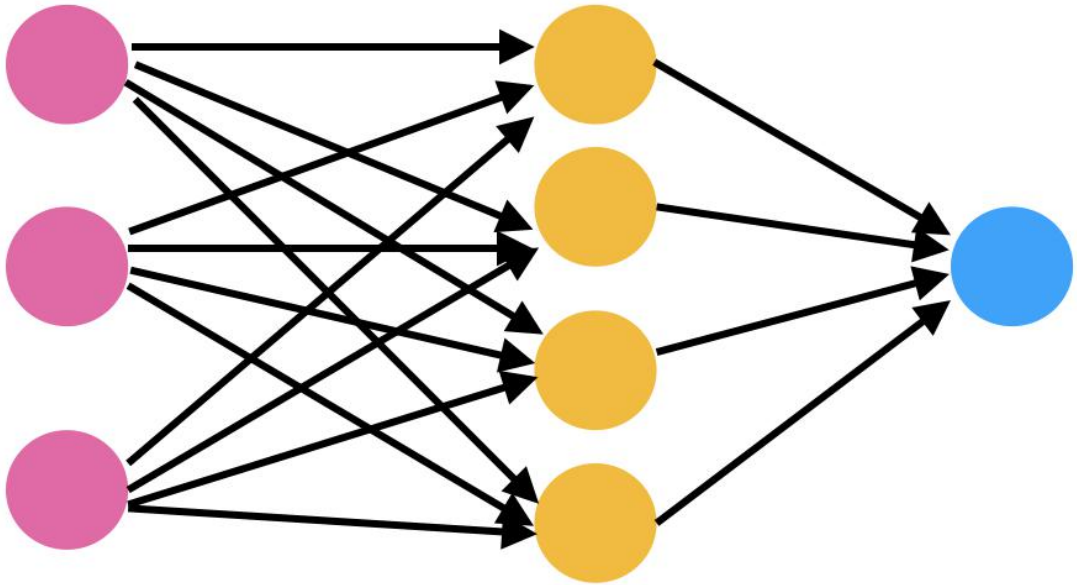
Another way is to impose non-linearity in our model $f_{\boldsymbol{\theta}}(\boldsymbol{x})$ (nonlinear in $\boldsymbol{\theta}$), leading to nonlinear supervised learning model.

- Note that supervised learning model is to use $f_{\boldsymbol{\theta}}(\boldsymbol{x})$ to approximate the underlying (nonlinear) pattern g.
- One most representative model is neural networks due to its universal approximation power.

$\rightsquigarrow$ Let us start with the basic model of neural networks.

Structure of Neural Network

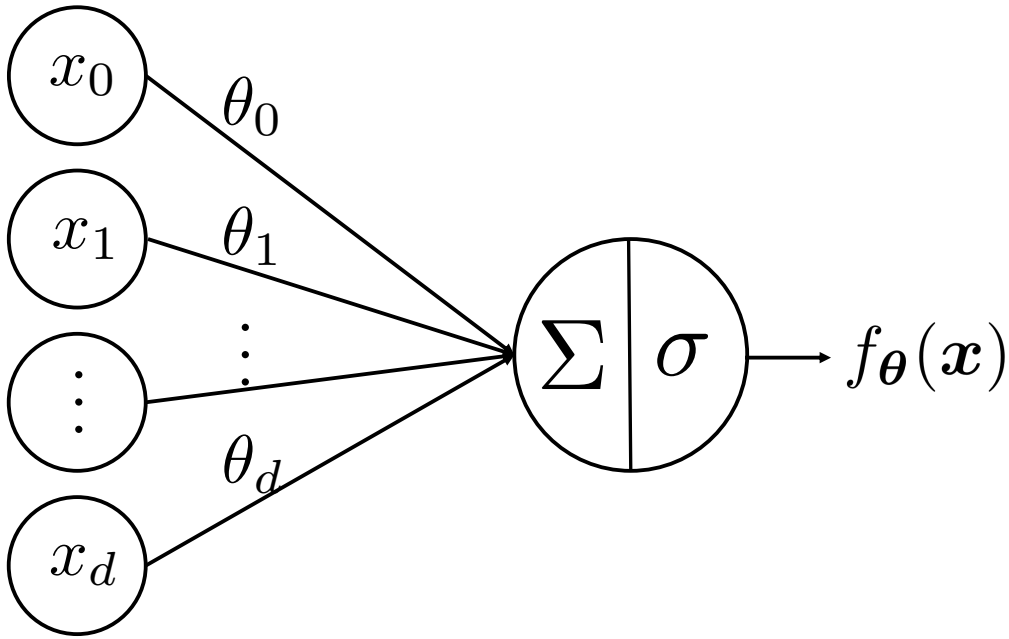
Neural network structure:



- Originally conceived as models for the brain.
  - Nodes are neurons.
  - Edges are synapses.

Let's start with the simplest single layer neural network.

Single Layer Neural Network

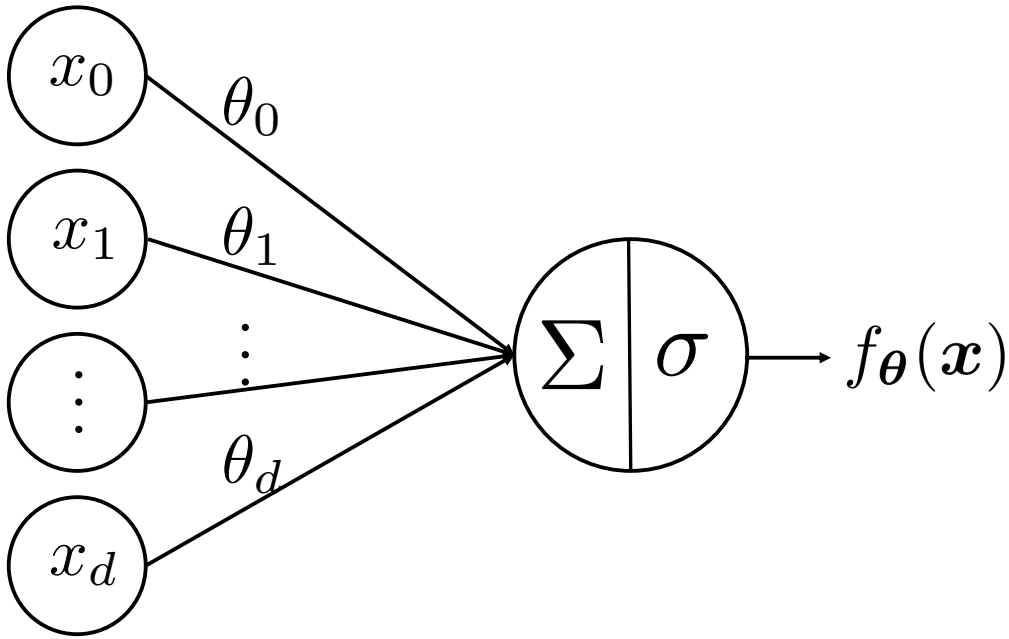


- Sample $\boldsymbol{x} = (x_0, \ldots, x_d)$ with $x_0 = 1$ (bias term).
- Parameters $\boldsymbol{\theta} = (\theta_0, \ldots, \theta_d)$, called weights.
- σ is called the activation function.

This neural network (NN) structure means

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = \sigma\left(\sum_{i=0}^{d} \theta_i x_i\right) = \sigma(\boldsymbol{\theta}^\top \boldsymbol{x}).$$

Neural Network As A Generalized Linear Model



This single layer neural network covers linear models as special cases.

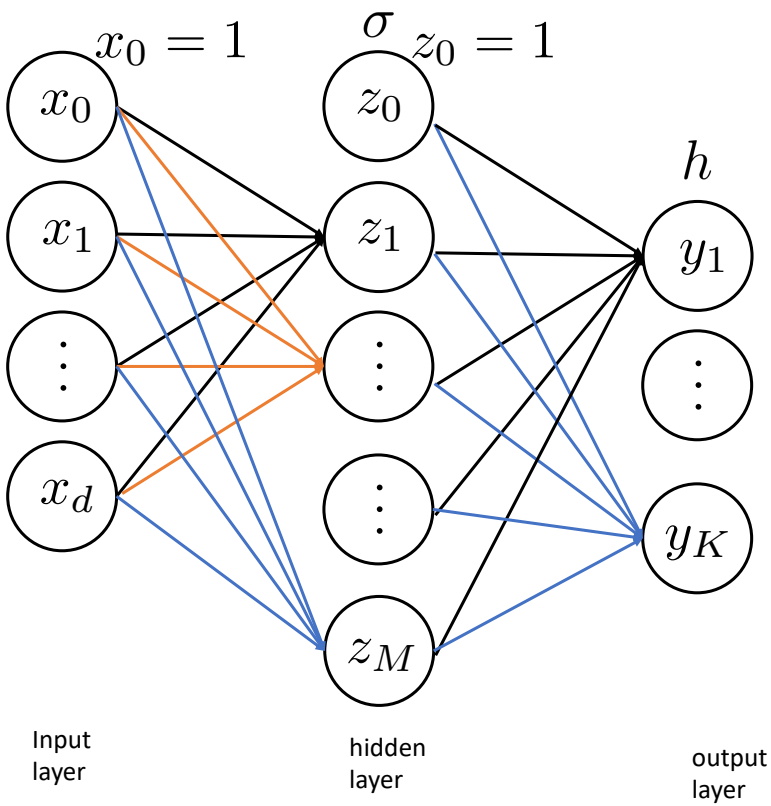
- Vanilla linear regression model if $\sigma(t) = t$.
- Logistic regression (binary case) if

$$\sigma(t) = \frac{1}{1 + e^{-y \cdot t}}$$

More generally, neural networks can have more than one layer.

One Hidden Layer (Two-Layer) Neural Network

We generalize neural network to two-layer:



Model:

$$z_m = \sigma(\boldsymbol{v}_m^\top \boldsymbol{x}), \quad m = 1, \ldots, M$$
$$y_k = h(\boldsymbol{w}_k^\top \boldsymbol{z}), \quad k = 1, \ldots, K$$

One Hidden Layer Neural Network

Mathematically, this one hidden layer neural network is represented as

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The main idea: Use the output of M linear models + possibly nonlinear transformation σ as the input to another linear model.

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$$\boldsymbol{v}_m, \quad \boldsymbol{w}_k, \qquad m = 1, \ldots, M \quad \text{and} \quad k = 1, \ldots, K$$

- What are the dimension of $\boldsymbol{v}_m$ and $\boldsymbol{w}_k$?
- How many extra parameters compared to vanilla linear model?

Neural Network Model in Matrix Form

$$z_m = \sigma(\boldsymbol{v}_m^\top \boldsymbol{x}), \quad m = 1, \ldots, M.$$
$$y_k = h(\boldsymbol{w}_k^\top \boldsymbol{z}), \quad k = 1, \ldots, K.$$

Let us omit the offset/bias for simplicity and without loss of generality. Letting $\boldsymbol{V} \in \mathbb{R}^{M \times d}$ denote the matrix having rows $\boldsymbol{v}_m^\top$ and $\boldsymbol{W} \in \mathbb{R}^{K \times M}$ denote the matrix having rows $\boldsymbol{w}_k^\top$, we can write this neural net as

$$\boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x}).$$
$$\boldsymbol{y} = h(\boldsymbol{W}\boldsymbol{z}).$$

One hidden layer NN model can be written compactly

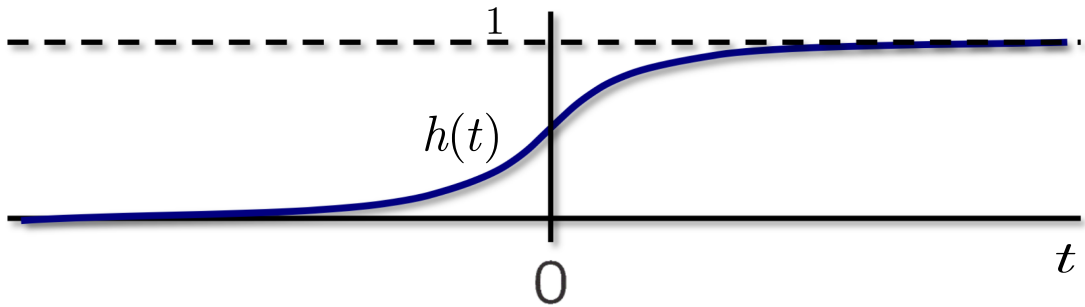
$$\boxed{\boldsymbol{y} = f_{\boldsymbol{\theta}}(\boldsymbol{x}) = h(\boldsymbol{W}\sigma(\boldsymbol{V}\boldsymbol{x})).}$$

- $\boldsymbol{\theta} := (\boldsymbol{V}, \boldsymbol{W})$ contains the network parameters, called weights.
- The input of the next layer is the output of the previous layer ($\boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x})$).

The Ingredients in Typical Neural Networks

- A historically common choice for σ is the logistic / sigmoid function:

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$



- A similar choice to sigmoid activation is tanh activation:

$$\sigma(t) = \frac{e^{t} - e^{-t}}{e^{t} + e^{-t}}.$$

- Another quite popular choice of σ is the ReLU (rectified linear unit) activation:

$$\sigma(t) = \max(0, t).$$

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The choice of h depends somewhat on the application.

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- Multiclass classification

$$h(t_k) = \frac{e^{t_k}}{\sum_{j=1}^{K} e^{t_j}}, \quad t_k = \boldsymbol{w}_k^\top \boldsymbol{z},$$

  which is to let the last layer to be a multi-class logistic regression classifier.

$\rightsquigarrow$ h is to apply certain linear model ($\boldsymbol{z}$ as input) at the end of NN.

The Representation Power of Neural Networks

One natural question would be why consider neural network model for nonlinear case.

- Supervised learning is to approximate the underlying pattern g. The target function g is nonlinear in general.
- The following theorem shows a universal approximation power of just one hidden layer neural networks, which shows that neural networks can represent any regular function, and thus can approximate any underlying (possibly nonlinear) g.

Theorem: Universal approximation Power

Let g be a continuous function on a bounded subset of d-dimensional space. Then, there exists a one hidden layer neural network $f_{\boldsymbol{\theta}}$ with a finite number of hidden neurons that approximates g arbitrarily well. Namely, for all sample $\boldsymbol{x}$, we have $|f_{\boldsymbol{\theta}}(\boldsymbol{x}) - g(\boldsymbol{x})| < \varepsilon$ for every $\varepsilon > 0$.

Remarks

- Like kernel methods, neural networks fit a linear model in a nonlinear feature space.
- Unlike kernel methods, these nonlinear features are learned — i.e., the output of the last hidden layer.
- Interpretation of NN: One can think neural network model as extracting features by nonlinear network and finally putting the extracted feature $\boldsymbol{z}$ into the last layer for regression or classification. So, the last layer can be viewed as a linear model such as LS, LR, SVM, etc.
  $\rightsquigarrow$ This is also true for deep learning, i.e., deep neural network model (later).
- We will see that the training of neural network model involves nonconvex optimization because of the involved nonlinear structure of $f_{\boldsymbol{\theta}}$.

Agenda

Training Neural Networks

- Training data: $(\boldsymbol{x}_1, \boldsymbol{y}_1), \ldots, (\boldsymbol{x}_n, \boldsymbol{y}_n)$ with $\boldsymbol{x}_i \in \mathbb{R}^d$ and $\boldsymbol{y}_i \in \mathbb{R}^K$
- Neural network model

$$f_{\boldsymbol{\theta}}(\boldsymbol{x}) = h(\boldsymbol{W}\sigma(\boldsymbol{V}\boldsymbol{x})).$$

- We aim to learn the weight parameters $\boldsymbol{\theta} = (\boldsymbol{V}, \boldsymbol{W})$ such that

$$\boldsymbol{y}_i \leftarrow f_{\boldsymbol{\theta}}(\boldsymbol{x}_i).$$

  Thus, this is a supervised learning problem.

- We can quantify this approximation by choosing a loss function which we will seek to minimize by picking $\boldsymbol{\theta}$ appropriately

The Learning Problem for Training Neural Networks

Regression: $h(t) = t$ and use squared error (ℓ_2) loss, resulting in

- $K = 1$

$$\min_{\boldsymbol{\theta}} \left\{ \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} (y_i - f_{\boldsymbol{\theta}}(\boldsymbol{x}_i))^2 \right\}$$

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Classification (Multi-class): $K > 1$, $\boldsymbol{y} = (0, \ldots, 1, \ldots, 0)$, h is softmax and use multiple-class logistic loss, resulting in

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Training Neural Networks is Nonconvex Optimization

For example, regression training problem can be written as:

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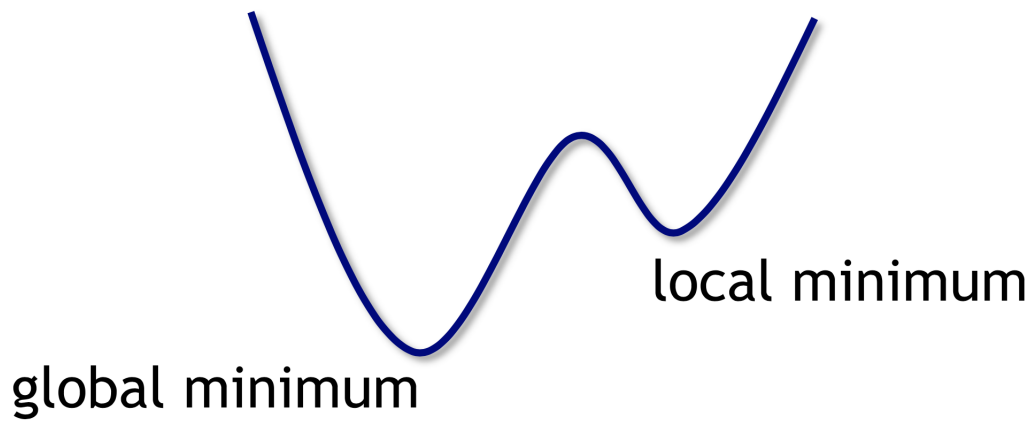

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We put an abstract form of the former learning problems:

$$\min_{\boldsymbol{\theta}} \left\{ \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \ell_i(\boldsymbol{\theta}) \right\}$$

For different applications (regression or classification), ℓ_i has its own form.

- One can apply a gradient-based training algorithm.
- One needs to compute the gradient

$$\nabla \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \nabla \ell_i(\boldsymbol{\theta}).$$

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$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \mu_k \nabla \mathcal{L}(\boldsymbol{\theta}_k).$$

However... The gradient is not easy to compute and GD training algorithm might be infeasible.

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Summary: NN learning formulation is the same as least squares or logistic regression, with the difference being using $\boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x})$ as input.

Training Neural Networks is Nonconvex Optimization

For example, regression training problem can be written as:

$$\min_{\boldsymbol{\theta}=(\boldsymbol{V},\boldsymbol{W})} \left\{ \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n} \sum_{i=1}^{n} \|\boldsymbol{y}_i - h(\boldsymbol{W}\sigma(\boldsymbol{V}\boldsymbol{x}_i))\|^2 \right\}.$$

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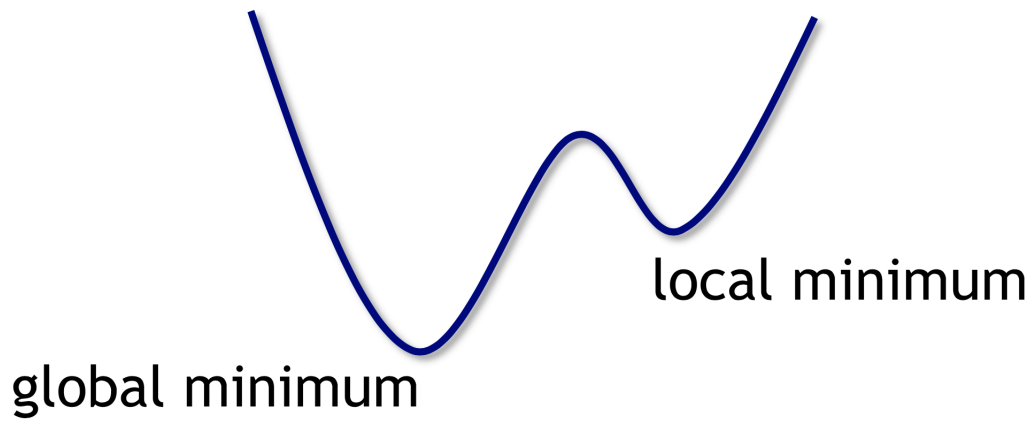

Recall that linear supervised learning always gives rise to convex optimization. The nonconvexity of NN learning comes from learning the feature $\boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x})$, where $\boldsymbol{V}$ is part of the learnable parameters.

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We put an abstract form of the former learning problems:

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However... The gradient is not easy to compute and GD training algorithm might be too optimistic.

Next: Training Algorithm and its Ingredients

We next dive into the depth of neural network training:

- How to compute the gradient?
  $\rightsquigarrow$ The well-known backpropagation (BP).
- In contemporary applications, n is so large. Applying GD is not feasible.
  $\rightsquigarrow$ Stochastic gradient descent, Adagrad, and Adam family.

Agenda

Computing The Gradient: Squared ℓ_2-Loss as An Example

- Let us take the regression case where we use squared error (ℓ_2) loss as an example. The conclusion applies to other cases.
- We consider the general case where $K > 1$. We have

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \|\boldsymbol{y}_i - f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\|^2 = \frac{1}{n}\sum_{i=1}^{n} \ell_i(\boldsymbol{\theta})$$

  where

$$\ell_i(\boldsymbol{\theta}) = \|\boldsymbol{y}_i - f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\|_2^2$$

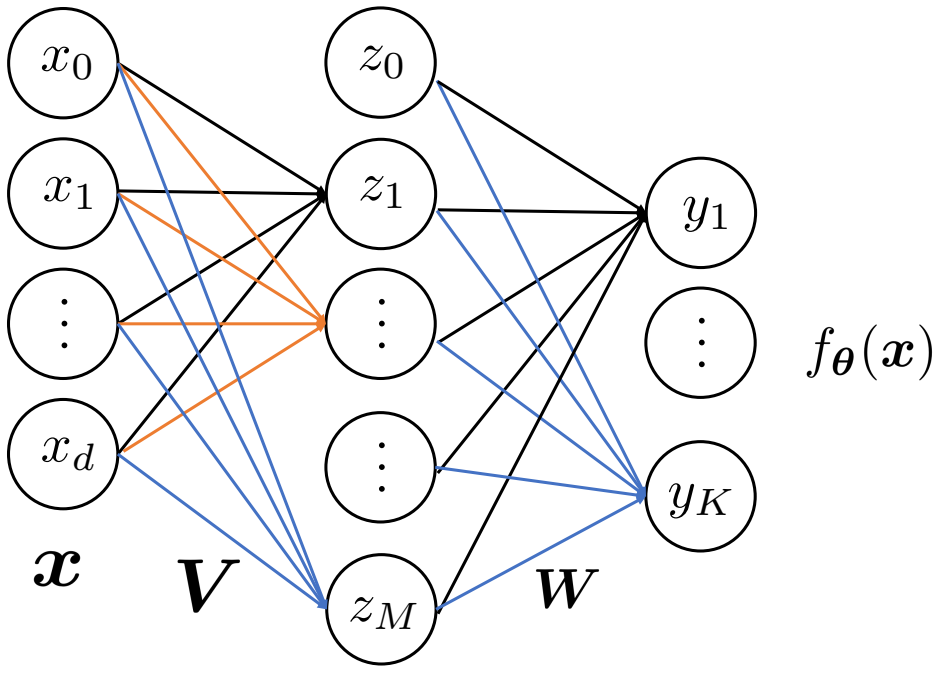
- Knowing how to take gradient for each ℓ_i suffices.

For ease of notation, we will omit the subscript i in ℓ_i and denote

$$\ell(\boldsymbol{\theta}) = \|\boldsymbol{y} - f_{\boldsymbol{\theta}}(\boldsymbol{x})\|^2.$$

Task: Computing $\nabla \ell(\boldsymbol{\theta}) = (\frac{\partial \ell}{\partial \boldsymbol{V}}, \frac{\partial \ell}{\partial \boldsymbol{W}})$.

Backpropagation for Gradient Computation: Overview



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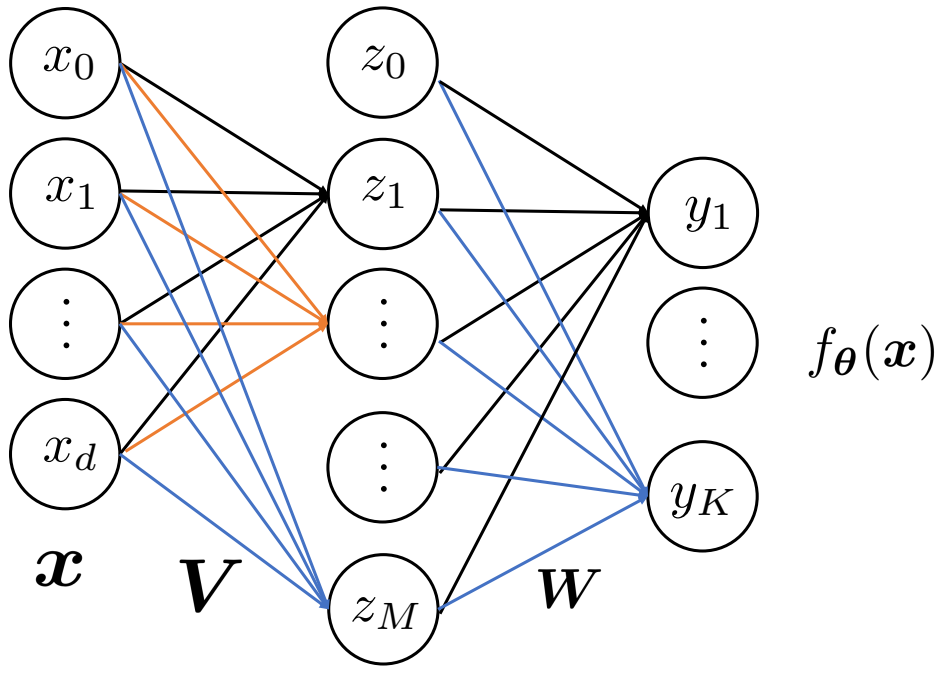


The idea: Computing gradient using chain rule:

- **Forward pass**: Given an $\boldsymbol{x}$, compute (using the current parameters $\boldsymbol{V}$ and $\boldsymbol{W}$) $\boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x}), f_{\boldsymbol{\theta}}(\boldsymbol{x}) = h(\boldsymbol{W}\boldsymbol{z}), \ell(\boldsymbol{\theta}) = \|\boldsymbol{y} - f_{\boldsymbol{\theta}}(\boldsymbol{x})\|_2^2$
- **Backward pass**:
  - Second layer: compute $\frac{\partial \ell}{\partial \boldsymbol{W}}$, and compute $\frac{\partial \ell}{\partial \boldsymbol{z}}$.
  - First layer: Compute $\frac{\partial \ell}{\partial \boldsymbol{V}} = \frac{\partial \ell}{\partial \boldsymbol{z}} \times \frac{\partial \boldsymbol{z}}{\partial \boldsymbol{V}}$.

Deriving The Gradient: Second Layer

- We omit the offset w.l.o.g. i.e., no x_0 and z_0.
- By definition of gradient, we have

$$\frac{\partial \ell}{\partial \boldsymbol{W}} = \begin{bmatrix} \frac{\partial \ell}{\partial W(1,1)} & \cdots & \frac{\partial \ell}{\partial W(1,M)} \\ \vdots & \ddots & \vdots \\ \frac{\partial \ell}{\partial W(K,1)} & \cdots & \frac{\partial \ell}{\partial W(K,M)} \end{bmatrix} \in \mathbb{R}^{K \times M}.$$

- We focus on one element $\frac{\partial \ell}{\partial W(k,m)}$ of $\frac{\partial \ell}{\partial \boldsymbol{W}}$ for simplicity.
- Note that during the forward pass, we have computed

$$\ell(\boldsymbol{\theta}) = \|\boldsymbol{y} - h(\boldsymbol{W}\boldsymbol{z})\|_2^2,$$

where $\boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x}) \in \mathbb{R}^M$.

Deriving The Gradient: Second Layer

$$
\begin{aligned}
\frac{\partial \ell}{\partial W(k,m)} &= \frac{\partial}{\partial W(k,m)} \|h(\boldsymbol{W}\boldsymbol{z}) - \boldsymbol{y}\|_2^2 \\
&= \frac{\partial}{\partial W(k,m)} \sum_{j=1}^{K} \left(h(\boldsymbol{w}_j^\top \boldsymbol{z}) - y[j]\right)^2 \\
&= \frac{\partial}{\partial W(k,m)} \left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right)^2 \qquad \text{using chain rule} \rightarrow \\
&= \frac{\partial \left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right)^2}{\partial \left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right)} \times \frac{\partial \left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right)}{\partial \boldsymbol{w}_k^\top \boldsymbol{z}} \times \frac{\partial \boldsymbol{w}_k^\top \boldsymbol{z}}{\partial W(k,m)} \\
&= 2\left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right) \times h'(\boldsymbol{w}_k^\top \boldsymbol{z}) \times z[m] \\
&:= \delta[k] \times z[m],
\end{aligned}
$$

where $\boldsymbol{w}_k^\top \in \mathbb{R}^M$ is the k-th row of $\boldsymbol{W}$ and we have defined

$$\delta[k] = 2\left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right) h'(\boldsymbol{w}_k^\top \boldsymbol{z}).$$

Deriving The Gradient: Second Layer

Since

$$\frac{\partial \ell}{\partial W(k,m)} = \delta[k] z[m]$$

Given $\delta[k] = 2\left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right) h'(\boldsymbol{w}_k^\top \boldsymbol{z})$, denote

$$\boldsymbol{\delta} = \begin{bmatrix} \delta[1] \\ \delta[2] \\ \vdots \\ \delta[K] \end{bmatrix} = 2(h(\boldsymbol{W}\boldsymbol{z}) - \boldsymbol{y}) \odot h'(\boldsymbol{W}\boldsymbol{z}) \in \mathbb{R}^K$$

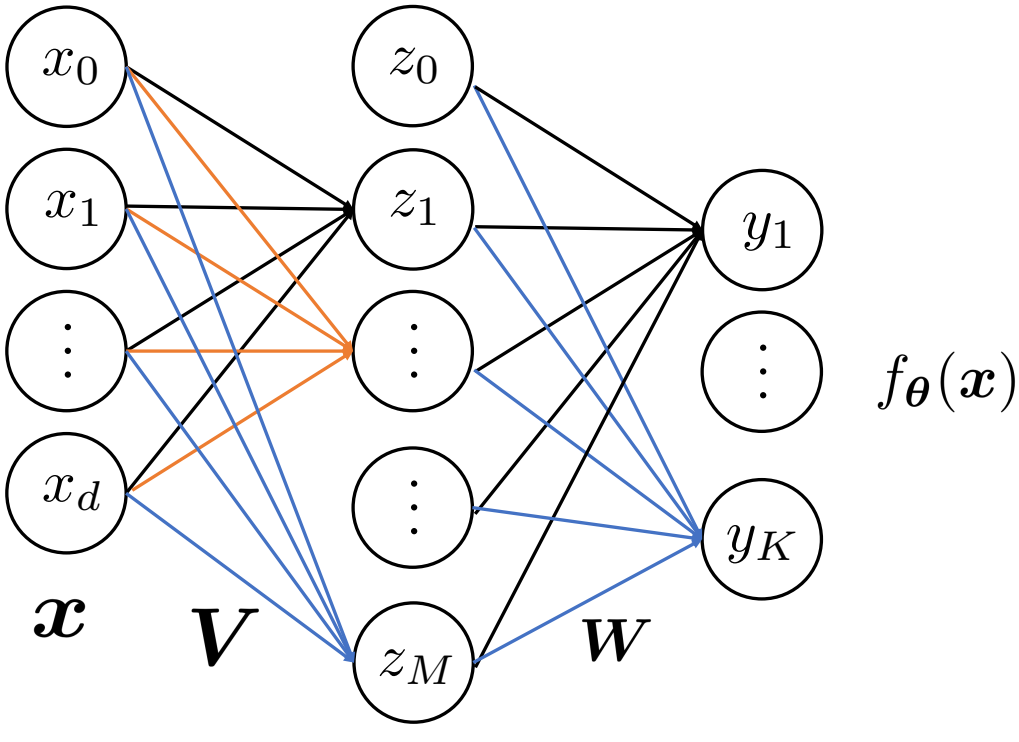
where $\odot$ is the Hadamard (element-wise) product.

We have

$$\frac{\partial \ell}{\partial \boldsymbol{W}} = \underbrace{\boldsymbol{\delta}}_{K\times 1} \underbrace{\boldsymbol{z}^\top}_{1\times M} \in \mathbb{R}^{K\times M},$$

which can be calculated using the current values of $\boldsymbol{W}$ and $\boldsymbol{z}$.

Deriving The Gradient: First Layer



$$\ell(\boldsymbol{\theta}) = \|\boldsymbol{y} - f_{\boldsymbol{\theta}}(\boldsymbol{x})\|_2^2, f_{\boldsymbol{\theta}}(\boldsymbol{x}) = h(\boldsymbol{W}\boldsymbol{z}), \boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x})$$

We firstly back-propagate the gradient back to $\boldsymbol{z}$, and then calculate the gradient of $\boldsymbol{z}$ w.r.t. $\boldsymbol{V}$

$$\frac{\partial \ell}{\partial \boldsymbol{V}} = \frac{\partial \ell}{\partial \boldsymbol{z}} \times \frac{\partial \boldsymbol{z}}{\partial \boldsymbol{V}}$$

Deriving The Gradient: First Layer

To begin with, we back-propagate the gradient back to $\boldsymbol{z}$.
For an individual node of $\boldsymbol{z}$:

$$\begin{aligned}
\frac{\partial \ell}{\partial \boldsymbol{z}[m]} &= \frac{\partial}{\partial \boldsymbol{z}[m]} \|h(\boldsymbol{W}\boldsymbol{z}) - \boldsymbol{y}\|_2^2 \\
&= \frac{\partial}{\partial \boldsymbol{z}[m]} \sum_{k=1}^{K} \left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right)^2 \quad \text{using chain rule} \rightarrow \\
&= \sum_{k=1}^{K} 2\left(h(\boldsymbol{w}_k^\top \boldsymbol{z}) - y[k]\right) \times h'(\boldsymbol{w}_k^\top \boldsymbol{z}) \times w_k[m] \\
&= \sum_{k=1}^{K} \delta[k] w_k[m]
\end{aligned}$$

Combining the individual nodes of $\boldsymbol{z}$, we denote

$$\frac{\partial \ell}{\partial \boldsymbol{z}} = \boldsymbol{W}^\top \boldsymbol{\delta}$$

Deriving The Gradient: First Layer

$$\frac{\partial \ell}{\partial \boldsymbol{V}} = \begin{bmatrix} \frac{\partial \ell}{\partial V(1,1)} & \cdots & \frac{\partial \ell}{\partial V(1,d)} \\ \vdots & \ddots & \vdots \\ \frac{\partial \ell}{\partial V(M,1)} & \cdots & \frac{\partial \ell}{\partial V(M,d)} \end{bmatrix} \in \mathbb{R}^{M \times d}.$$

For an individual element of $\boldsymbol{V}$:

$$\frac{\partial \ell}{\partial V(m,j)} = \underbrace{\frac{\partial \ell}{\partial \boldsymbol{z}[m]}}_{\text{gradient from next layer}} \times \underbrace{\frac{\partial \boldsymbol{z}[m]}{\partial V(m,j)}}_{\text{local gradient}} \qquad \text{(chain rule)}$$

$$= \frac{\partial \ell}{\partial \boldsymbol{z}[m]} \times \frac{\partial \sigma(\boldsymbol{v}_m^\top \boldsymbol{x})}{\partial V(m,j)} \qquad (\text{since } \boldsymbol{z} = \sigma(\boldsymbol{V}\boldsymbol{x}))$$

$$= \sum_{k=1}^{K} \delta[k] w_k[m] \times \sigma'(\boldsymbol{v}_m^\top \boldsymbol{x}) \times x[j]$$

Combining the gradients of the individual elements:

$$\frac{\partial \ell}{\partial \boldsymbol{V}} = \underbrace{\left((\boldsymbol{W}^\top \boldsymbol{\delta}) \odot \sigma'(\boldsymbol{V}\boldsymbol{x}) \right)}_{M \times 1} \times \underbrace{\boldsymbol{x}^\top}_{1 \times d} \in \mathbb{R}^{M \times d}$$

The Overall Procedure

- Recall that for each data $\boldsymbol{x}_i$, we need to compute its corresponding gradient $\nabla \ell_i(\boldsymbol{\theta})$.
- Since $\boldsymbol{\theta} = (\boldsymbol{V}, \boldsymbol{W})$, we have

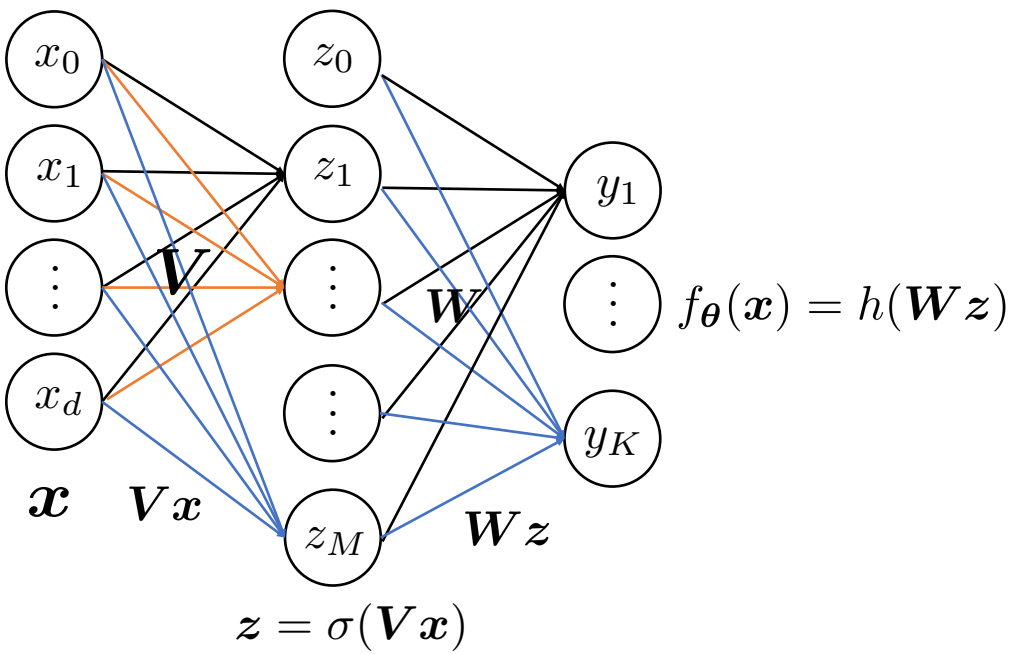
$$\nabla \ell_i(\boldsymbol{\theta}) = \left(\frac{\partial \ell_i}{\partial \boldsymbol{V}}, \frac{\partial \ell_i}{\partial \boldsymbol{W}} \right)$$

- This is achieved by an efficient two-pass method for computing the gradient at each iteration—the so-called backpropagation:
  1. Forward pass
  2. Backward pass

Backpropagation: Forward pass

Forward pass: Feed $\boldsymbol{x}_i$ into the network, use the current parameters $\boldsymbol{V}$ and $\boldsymbol{W}$ to compute and store

- $\boldsymbol{V}\boldsymbol{x}_i$
- $\boldsymbol{z}_i = \sigma(\boldsymbol{V}\boldsymbol{x}_i)$
- $\boldsymbol{W}\boldsymbol{z}_i$
- $f_{\boldsymbol{\theta}}(\boldsymbol{x}_i) = h(\boldsymbol{W}\boldsymbol{z}_i)$, $\ell_i(\boldsymbol{\theta}) = \|\boldsymbol{y} - f_{\boldsymbol{\theta}}(\boldsymbol{x}_i)\|_2^2$



Backpropagation: Backward pass

Backward pass:

Compute the gradient of the second layer

- $\boldsymbol{\delta}_i = 2(h(\boldsymbol{W}\boldsymbol{z}_i) - \boldsymbol{y}_i) \odot h'(\boldsymbol{W}\boldsymbol{z}_i)$
- $\frac{\partial \ell_i}{\partial \boldsymbol{W}} = \boldsymbol{\delta}_i \boldsymbol{z}_i^\top$

Backpropagate the gradient to $\boldsymbol{z}_i$:

- $\frac{\partial \ell}{\partial \boldsymbol{z}_i} = \boldsymbol{W}^\top \boldsymbol{\delta}_i$

Compute the gradient of the first layer:

- $\frac{\partial \ell_i}{\partial \boldsymbol{V}} = \left(\frac{\partial \ell}{\partial \boldsymbol{z}_i} \odot \sigma'(\boldsymbol{V}\boldsymbol{x}_i) \right) \boldsymbol{x}_i^\top$

Thus, backpropagation is an efficient way of computing the gradient of NN learning problem.

Given the computed gradient, we can apply gradient-based training algorithm for training NN. $\rightsquigarrow$ Which algorithm should we choose?

Structured Learning Problem: Finite-Sum Structure

In LS, LR, SVM, and NN, we always have

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n} \underbrace{\ell(\boldsymbol{\theta}; \boldsymbol{x}_i, y_i)}_{\ell_i(\boldsymbol{\theta})}$$

each $\ell_i(\boldsymbol{\theta})$ is called a component function. We can use BP to compute $\nabla\mathcal{L}(\boldsymbol{\theta})$ if it is NN.

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Example: Least squares

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^d} \ \frac{1}{n}\|\boldsymbol{X}\boldsymbol{\theta} - \boldsymbol{y}\|_2^2 = \frac{1}{n}\sum_{i=1}^{n} \underbrace{(\boldsymbol{\theta}^\top \boldsymbol{x}_i - y_i)^2}_{\ell_i(\boldsymbol{\theta})}$$

- Each ℓ_i corresponds to one training data point $(\boldsymbol{x}_i, y_i)$.
- The above structure is called finite-sum and the problem is called finite-sum optimization.
- Finite-sum optimization is ubiquitous in machine learning, including most of the (supervised) learning problems.

Modern Large-Scale Machine Learning Problems

One of the key features of modern machine learning problems is large-scale, i.e., the number of training data n is very large (guided by VC analysis). Such a problem is called large-scale machine learning.

- One notable example is training NN.

We have the following two observations:

- The property we want for an algorithm is fast convergence. In terms of optimization, we should use fast algorithm (fast in terms of iteration) like first-order method (GD/AGD) or even second-order method (Newton).
- Guided by VC analysis, we just need to choose a fine $\mathcal{H}$ with small generalization error and then design a fast learning algorithm (GD/AGD or Newton) to learn $\widehat{\boldsymbol{\theta}}$ by solving the learning problem. Then, the learning process is done.

What is missed?

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What is missed? We do not consider **the computational issue** of the learning algorithm when n is large-scale.

Let us take GD as an example.

Gradient Descent in Large-scale Finite-sum Structure

GD:

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\theta}_k - \mu_k \nabla \mathcal{L}(\boldsymbol{\theta}_k)$$

- What is the structure of $\nabla \mathcal{L}(\boldsymbol{\theta}_k)$?

Gradient Descent in Large-scale Finite-sum Structure

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- What is the structure of $\nabla \mathcal{L}(\boldsymbol{\theta}_k)$?

Due to the linearity of gradient operator, we have

$$\begin{aligned} \nabla \mathcal{L}(\boldsymbol{\theta}_k) &= \nabla \left(\frac{1}{n} \sum_{i=1}^{n} \ell_i(\boldsymbol{\theta}_k) \right) \\ &= \frac{1}{n} \sum_{i=1}^{n} \nabla \ell_i(\boldsymbol{\theta}_k) \end{aligned}$$

- Observation: The gradient of $\mathcal{L}$ is the summation of the gradients of all the component functions.
- It can be costly and even prohibitive to the compute the full gradient when n is very large.
- Newton's method is even worse when n is so large.